

Grade 6 ICT Notes – Using the Computer Laboratory Safely

1. Computer Laboratory

A **computer laboratory** is a special room equipped with computers and other devices such as printers, scanners, multimedia projectors, and networking equipment.

These devices are **valuable and expensive**, so everyone must use them carefully and responsibly.

2. Safety in the Computer Laboratory

A. Guidelines for Your Safety

- Be careful when touching electrical wires and sockets.
 - Do not run or play inside the laboratory.
 - Know where the emergency exit doors are located.
 - Be aware of fire extinguishers in the laboratory.
 - Always follow the teacher's instructions.
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B. Guidelines for the Safety of Equipment

- Do not switch on any device without permission.
 - Switch off all equipment properly after use.
 - Scan external storage devices (USB drives, external hard disks) for viruses before using them.
 - Return equipment to its proper place after use.
 - Do not bring food, drinks, or liquids into the laboratory.
 - Remove footwear if required by laboratory rules.
 - Keep the laboratory and equipment clean and dust-free.
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3. Starting a Computer

Follow these steps when turning on a desktop computer.

1. Switch on the wall power supply.
2. Turn on the UPS (Uninterruptible Power Supply), if available.
3. Switch on the System Unit.
4. Turn on the Monitor.

Note: Modern computers and laptops may have only one power button.

Always follow your teacher's instructions because the procedure may differ.

4. Shutting Down a Computer

Always shut down the computer properly.

Steps

1. Save your work.
2. Close all open programs.
3. Click **Shutdown** from the Start Menu.
4. Wait until the monitor screen becomes dark.
5. Turn off the monitor.
6. Turn off the UPS.
7. Switch off the wall power supply.

Why is proper shutdown important?

- Protects computer files.
 - Prevents data loss.
 - Increases the life of the computer.
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5. Computer Ethics

Computer Ethics are the rules and good practices that should be followed while using computers.

Good Computer Ethics

- Use the computer without disturbing others.
 - Do not copy or use paid software illegally.
 - Do not repair computer equipment without permission.
 - Do not access the Internet without teacher supervision.
 - Do not uninstall, change, or delete programs without permission.
 - Never hack or interfere with another person's computer.
 - Do not use another person's computer without permission.
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6. Correct Posture While Using a Computer

Maintaining the correct posture helps prevent health problems.

Correct Posture

A. Screen Position

- Keep the monitor at eye level or slightly below.

B. Viewing Distance

- Keep the monitor **18–28 inches (45–70 cm)** away from your eyes.

C. Sitting Position

- Sit upright with your back supported by the chair.

D. Leg Position

- Keep your feet flat on the floor.

E. Keyboard and Mouse Position

- Keep the keyboard and mouse at elbow level.
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Health Problems Caused by Incorrect Posture

Incorrect Practice

Possible Health Problem

Looking too closely at the screen Eye strain, blurred vision, watery eyes

Incorrect Practice	Possible Health Problem
Sitting with a bent back	Back pain
Poor leg position	Leg strain
Incorrect keyboard height	Wrist and arm discomfort

Additional Tips

- Adjust screen brightness and contrast.
- Rest your eyes regularly during long computer sessions.

7. Electronic Waste (E-Waste)

Electronic Waste (E-Waste) refers to old, damaged, or unused electronic devices.

Examples:

- Old computers
- Keyboards
- Monitors
- Printers
- Mobile phones

E-waste contains harmful materials such as:

- Lead
- Mercury
- Beryllium
- Barium

These substances can harm both people and the environment.

8. The 3R Method of Managing E-Waste

A. Reduce

Use electronic devices carefully so they last longer.

Example:

- Repair a computer instead of buying a new one.
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B. Reuse

Use old equipment again if it still works.

Examples:

- Donate old computers.
 - Sell working computers.
 - Use old computer parts for other purposes.
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C. Recycle

If equipment cannot be repaired or reused:

- Send it to an authorized electronic waste recycling center.
 - Recycling converts old materials into new products.
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9. Password Protection

A **password** is a secret word or combination of characters used to protect a computer and its information.

Why Use Passwords?

- Prevent unauthorized access.
 - Protect personal information.
 - Keep files safe.
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Rules for Creating a Strong Password

A good password should:

- Contain at least **8 characters**.
- Include:
 - Uppercase and lowercase letters (A–Z, a–z)
 - Numbers (0–9)
 - Symbols (@, #, \$, %, etc.)
- Avoid using:
 - Your name
 - Birthday
 - Simple words
- Use a password hint if needed.
- Add a recovery email address to reset your password if forgotten.